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// **Question 1:**

```
#include <iostream>
#include <iomanip>
using namespace std;

int main() {
    double length, breadth, costPerSqMeter;

    cout << "Enter the length of the room (in meters): ";
    cin >> length;

    while (length <= 0) {
        cout << "Error: Length must be a positive value. Please enter
again: ";
        cin >> length;
    }

    cout << "Enter the breadth of the room (in meters): ";
    cin >> breadth;

    while (breadth <= 0) {
        cout << "Error: Breadth must be a positive value. Please enter
again: ";
        cin >> breadth;
    }

    cout << "Enter the cost per square meter of the carpet: ";
    cin >> costPerSqMeter;

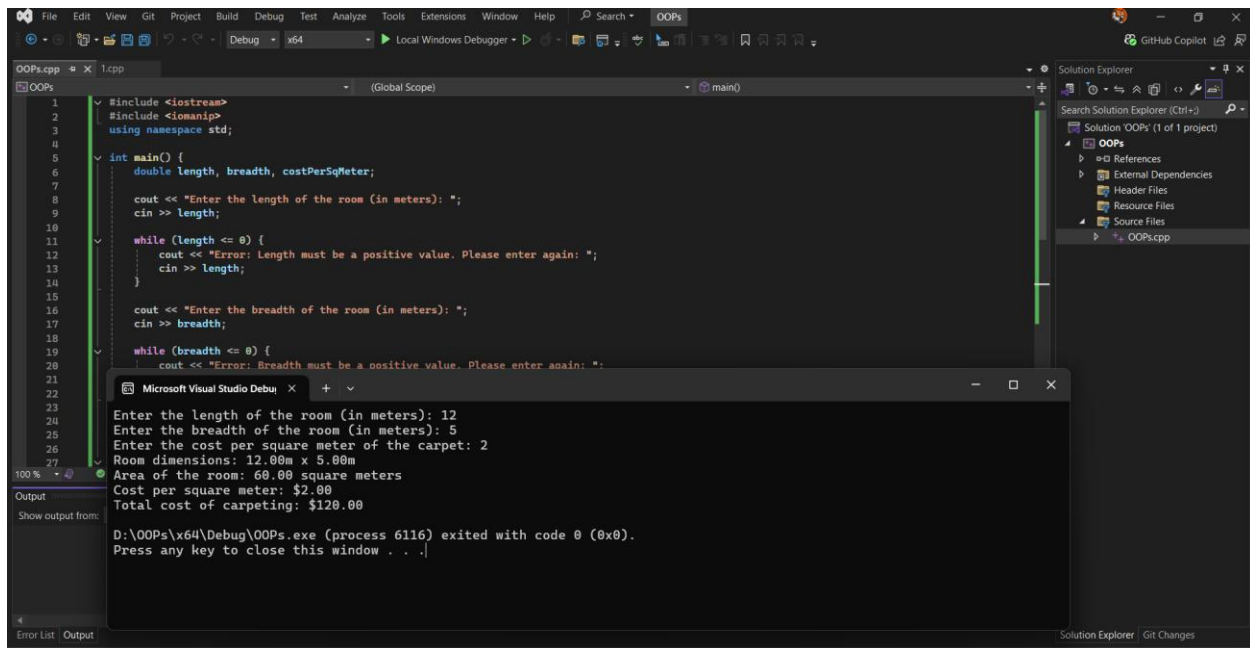
    while (costPerSqMeter <= 0) {
        cout << "Error: Cost must be a positive value. Please enter again:
";
        cin >> costPerSqMeter;
    }

    double area = length * breadth;

    double totalCost = area * costPerSqMeter;

    cout << fixed << setprecision(2);
    cout << "Room dimensions: " << length << "m x " << breadth << "m" <<
endl;
    cout << "Area of the room: " << area << " square meters" << endl;
    cout << "Cost per square meter: $" << costPerSqMeter << endl;
    cout << "Total cost of carpeting: $" << totalCost << endl;

    return 0;
}
```



//Question 2:

```
#include <iostream>
#include <cstdlib>
#include <ctime>
#include <limits>
using namespace std;

int main() {
    srand(time(0));
    int actualNumber = rand() % 100 + 1;
    int guess;
    int trials = 3;

    cout << "I'm thinking of a number between 1 and 100." << endl;
    cout << "You have " << trials << " trials to guess it!" << endl <<
endl;

    for (int attempt = 1; attempt <= trials; attempt++) {
        cout << "Trial " << attempt << ": Enter your guess (1-100): ";

        // Input validation
        while (!(cin >> guess) || guess < 1 || guess > 100) {
            cin.clear();
            cin.ignore(numeric_limits<streamsize>::max(), '\n');
            cout << "Invalid input! Please enter a number between 1 and
100: ";
        }
    }
}
```

```

        if (guess == actualNumber) {
            cout << "Congratulations! You guessed the correct number: " <<
actualNumber << endl;
            break;
        }
        else if (guess > actualNumber) {
            cout << "Your guess is HIGHER than the actual number." <<
endl;
        }
        else {
            cout << "Your guess is LOWER than the actual number." << endl;
        }

        if (attempt == trials) {
            cout << "Game over! You've used all your trials." << endl;
            cout << "The actual number was: " << actualNumber << endl;
        }
        else {
            cout << "Try again!" << endl << endl;
        }
    }

    return 0;
}

```

The screenshot displays the Microsoft Visual Studio IDE. The main editor window shows the C++ source code for a number-guessing game. The console window at the bottom shows the program's execution. The output indicates that the user has 3 trials to guess a number between 1 and 100. The user enters 75, 50, and 25. The program responds with 'Your guess is HIGHER than the actual number.' for the first two trials and 'Your guess is LOWER than the actual number.' for the third. After the third trial, it says 'Game over! You've used all your trials. The actual number was: 45'. The program then exits with code 0. The bottom panel shows the build output, including a warning C4244 and a message that the build succeeded.

```

I'm thinking of a number between 1 and 100.
You have 3 trials to guess it!
Trial 1: Enter your guess (1-100): 75
Your guess is HIGHER than the actual number.
Try again!
Trial 2: Enter your guess (1-100): 50
Your guess is HIGHER than the actual number.
Try again!
Trial 3: Enter your guess (1-100): 25
Your guess is LOWER than the actual number.
Game over! You've used all your trials.
The actual number was: 45
D:\OOPs\x64\Debug\OOPs.exe (process 2400) exited with code 0 (0x0).
Press any key to close this window . . .

```

Build Output:

```

1>OOPs.cpp
1>OOPs.vcxproj -> D:\OOPs\x64\Debug\OOPs.exe
1>Done Building project "OOPs.vcxproj".
***** Build: 1 succeeded, 0 failed, 0 up-to-date, 0 skipped *****
***** Build completed at 16:24 and took 04.390 seconds *****

```

// Question 3

```
#include <iostream>
using namespace std;

int main() {
    int n;

    cout << "Enter a positive integer: ";
    cin >> n;

    while (n <= 0) {
        cout << "Error: Please enter a positive integer: ";
        cin >> n;
    }

    cout << "Sequence generated from " << n << ":" << endl;
    cout << n << " ";
    int count = 1;
    while (n != 1) {
        if (n % 2 == 0) {
            n = n / 2;
        }
        else {
            n = n * 3 + 1;
        }

        cout << n << " ";
        count++;
        if (count % 10 == 0) {
            cout << endl;
        }
    }
    cout << endl << "Sequence length: " << count << " numbers" << endl;
    return 0;
}
```

